Higher National Diploma in Engineering

2nd Year, First Semester Examination - 2018

CE2109-Soil Mechanics & Engineering Geology

Question 1 (20 Marks)

I. Draw a detailed three phase diagram of soil and derive the following equations: (08 Marks)

a) $n = e/(1+e)$
b) $\gamma_d = G_s\gamma_w/(1+e)$
c) $\gamma_{sat} = (G_s+e)\,\gamma_w/(1+e)$

II. A layer of silty sand, **5m in thickness** overlies a **5m thick bed of gravel**. The **ground water table is 2m below the ground surface**. Following data is given:

- Dry unit weight of sand =16.7kN/m^3
- Saturated unit weight of sand =19.1kN/m^3
- Saturated unit weight of gravel =20.1kN/m^3

Determine the **vertical total stress, pore water pressure, and vertical effective stress** at the depth of **10m** and plot the stress distribution diagrams (12 Marks).

Question 2 (20 Marks)

I. List five factors affecting the compaction (05 Marks).

II. Explain the principles of compaction (05 Marks).

III. Following results were obtained from a Proctor compaction test carried out on a soil. Obtain an estimate of the optimum moisture content and maximum dry density (10 Marks).

Moisture content (%)	**Bulk unit weight (kN/m3)**
8.5	18
9.7	20.1

Moisture content (%)	Bulk unit weight (kN/m3)
11.1	21.6
13.0	22.2
14.8	21.4
16.5	20.0

Question 3 (20 Marks)

I. State the Darcy's law for flow of water through saturated soils (02 Marks).

II. List four factors which are affecting permeability (04 Marks).

III. Using falling head method show that (08 Marks). $K = \frac{2.303}{At}\ aL\log(h_0/h_1)$

Where:

- k- coefficient of permeability
- a - cross sectional area of stand pipe
- L - length of soil sample
- A - cross sectional area of soil sample
- H0 - initial head
- H1 - final head
- t - duration of the test
- Hint: ln(x)=2.303 log(x)

IV. Following values were observed during a falling head permeability test. Determine the **coefficient of permeability** of the soil sample (06 Marks).

- Length of soil sample L=150mm
- Stand pipe diameter d=5mm

- Initial head of water in stand pipe =1500mm
- Final head of water in stand pipe =600mm
- Diameter of soil specimen=100mm
- Duration of test t=300 seconds

Question 4 (20 Marks)

I. Explain the importance of sieve analysis (05 Marks).

II. List the possible errors in sieve analysis (05 Marks).

III. The results of a sieve analysis test are shown in the table below (10 Marks).

Sieve no	**8**	**14**	**28**	**35**	**48**	**100**	**200**	**pan**
Diameter (mm)	2.362	1.168	0.580	0.417	0.295	0.147	0.074	--
Weight retained (g)	48.0	67.2	106.8	54.0	26.4	21.6	69.6	80

Total weight of the sample was **424g**. Plot the **particle size distribution curve** on the provided semi log graph sheet. Also determine the **coefficient of uniformity** and **coefficient of curvature**.

Question 5 (20 Marks)

Answer the following questions:

I. Briefly explain the three main types of rock and its formation.

II. Compare the advantages and limitations of direct shear test over triaxial test.

III. Briefly explain the active and passive earth pressure with neat sketch.

IV. Explain the importance of site investigation / exploration.

V. Define the following terms: fault, fold, strike & dip.

Question 6 (20 Marks)

I. A sample of soil was compacted in to a cylindrical mould of 10cm diameter and 15cm height. The weight of the moist sample was 2360g. After oven drying the weight of the sample was 2200g. Determine the following parameters of the soil (10 Marks):

a) Moisture content

b) Bulk density

c) Dry density

d) Voids ratio

e) Degree of saturation

II. Describe the laboratory procedure to determine LL (05 Marks).

III. Results of a LL test are shown in the table below. Determine the LL and PI of the tested soil (05 Marks).

No of blows	**31**	**27**	**22**	**18**
Moisture content (%)	31	33.5	37	39.5